

The empirical recurrence for OEIS A230802: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A230802 records “Empirical recurrence of order 68”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 71$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 68, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 68 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A230802 is “Number of $n \times 4$ 0..3 arrays $x(i,j)$ with each element horizontally or vertically next to at least one element with value $(x(i,j)+1) \bmod 4$ and at least one element with value $(x(i,j)-1) \bmod 4$, and upper left element zero.”. It has offset 1 and begins

0, 16, 40, 1370, 10448, 183088, 2045378, 29175688, ...

The entry states:

Empirical recurrence of order 68 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:02:09, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 30785$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 71$ of the 30785, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 71$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 142$ exact terms of the model returns order 68 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{68} c_i a(n-i),$$

c_1	28	c_{18}	−59275000682	c_{35}	−147706435063709	c_{52}	6065857725616
c_2	−228	c_{19}	113981876255	c_{36}	150097412708656	c_{53}	−3300437795231
c_3	−203	c_{20}	92933765424	c_{37}	−97939222547866	c_{54}	108444864185
c_4	13254	c_{21}	−717790751479	c_{38}	22838532619451	c_{55}	289046590142
c_5	−74558	c_{22}	1120216155814	c_{39}	57043198452457	c_{56}	−183975227324
c_6	84842	c_{23}	250965046403	c_{40}	−128488347427537	c_{57}	340269382184
c_7	813360	c_{24}	−3968111528441	c_{41}	156248560070040	c_{58}	−166782224192
c_8	−3790112	c_{25}	6749920473125	c_{42}	−115275658398972	c_{59}	−45181051608
c_9	2009849	c_{26}	−2312063123079	c_{43}	44349225635679	c_{60}	9829119696
c_{10}	34612982	c_{27}	−11035729666628	c_{44}	−7631978996085	c_{61}	56997480912
c_{11}	−107273468	c_{28}	22448019188495	c_{45}	7776293930871	c_{62}	−41379565040
c_{12}	−58906944	c_{29}	−13172259504084	c_{46}	−2202160412873	c_{63}	11440576832
c_{13}	968863483	c_{30}	−23181754623286	c_{47}	−14138161642072	c_{64}	−1076985152
c_{14}	−1414122504	c_{31}	63260971847814	c_{48}	14274971670759	c_{65}	−244179712
c_{15}	−3660774183	c_{32}	−65263557072290	c_{49}	317940431226	c_{66}	168278016
c_{16}	14426432375	c_{33}	7740121629829	c_{50}	−5784372981681	c_{67}	−41808896
c_{17}	−5243580787	c_{34}	82359892966310	c_{51}	−1348137764523	c_{68}	3399680

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 68, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $68 + 4$ terms removed returns order 68 as well, so $\deg q_{\min} = 68 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $68 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 68 that this sequence satisfies — and that family turns out to have exactly one member.

References

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